# e-Game Board Project

## Academic Year 2026/2027

**Project Overview**

The objective of this project is to develop an electronic board game in which each player is represented by a meeple on the game board and owns a control base that displays the player's current game status.

Each student group must define and implement the rules of the game. Each group will manage three meeples.

The game consists of two main components:

- **Meeple**
  - The meeple represents the player's position on the game board.
  - The meeple must be capable of detecting its position, reporting movement between board cells and determining whether it is the active meeple during the current turn.

  **Components:**

  - ESP-01 - Will oversee the meeple control logic.
  - LED - It flashes when it is the player's turn.
  - RFID Reader - Identifies the location and movement of the meeple on the game board (each cell has a RFID tag)

  Optional features: attack and defense mechanics, interactions between meeples and additional game-specific functionality.

- **Control base**
  - The control base displays player status and provides interaction mechanisms.
  - It identifies the active meeple, determines available movements, displays status information and generates sounds associated with game events.

  **Components:**

  - ESP32/FreeRTOS - It will oversee displaying and updating the players status.
  - LCD screen and a LED to show the player's tactical situation (health, remaining movements, etc.).
  - Buzzer - It indicates the player's action/status through sounds.
  - Push button - Allows the player interaction and the voluntary turn end.

**Game Controller - Player interaction**

The game components will interact through the MQTT protocol. To do this, it is required:

1. Identify each player uniquely

   When players join the game, they will receive an identification token that will be used during the game.

2. Each player component will interact exclusively through the MQTT protocol, subscribing and publishing to its topic/data so that:
   - The meeple will publish its movement and it subscribes to the necessary topic to receive its status (active turn or not) and any other meeples interaction defined by the game.
   - The control base will publish the remaining meeple movements, their health/status and it subscribes to the necessary topics to receive any other situations defined by the game.
3. The topics will be dependent on the rules of the game.

The central game controller runs on a PC and may be implemented in Python, C, C++, Lua or any suitable language.

**General project information**

- Groups consist of four students.
- The game board must support three simultaneous players.

**Project development and requirements**

1. Meeple player

- Hardware: ESP-01, LED, RFID reader
- Actions:
    - Must detect position changes.
    - Uses the MQTT protocol to transfer and receive the player status.
- Required:
    - Correctly identifies the change in location of the meeple.
    - It must connect to the Wi-Fi LAN.
    - The device acts as a MQTT client.

2. Operational Base station

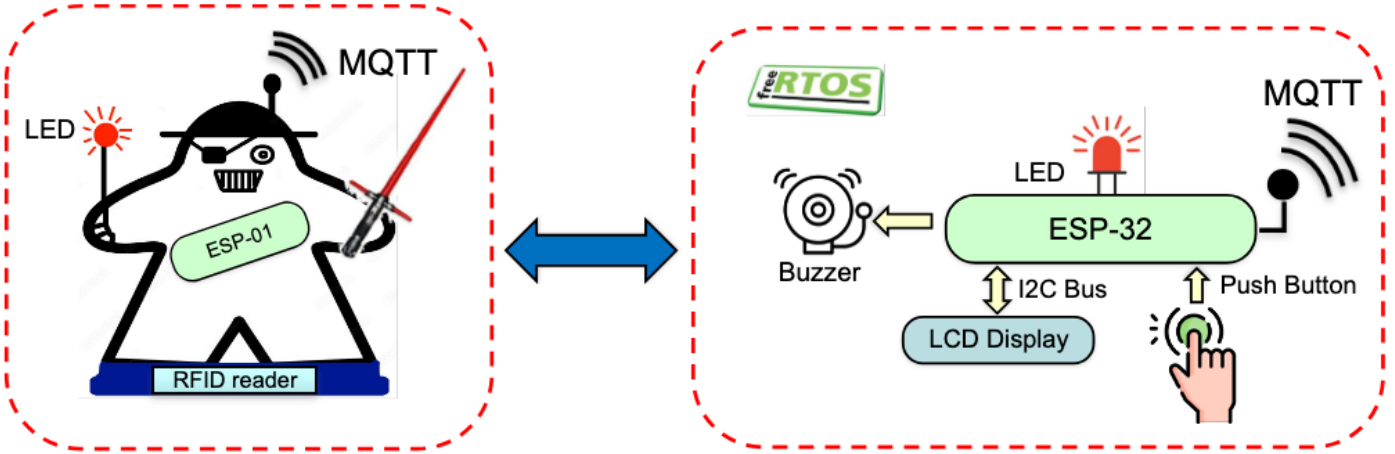
- Hardware: ESP32, LCD Screen, LED, Buzzer, Push Button,
- Actions:
    - Obtains the player status ans shows it in the LCD screen.
    - Use the buzzer to signal changes in the meeple's state.
    - Uses the MQTT protocol to transfer and receive the player status.
- Required:
    - Use of the FreeRTOS to implement the device functionalities.
    - It must connect to the Wi-Fi LAN.
    - The device acts as a MQTT client.

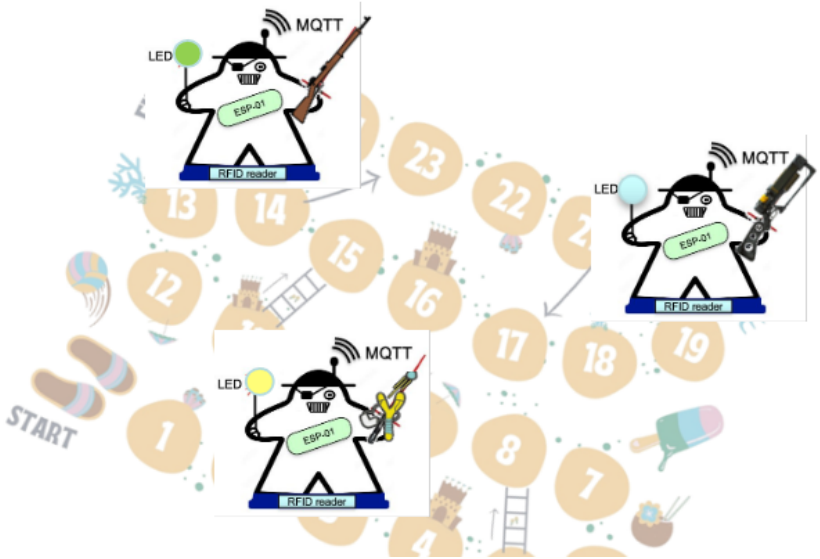
3. MQTT Broker

- Hardware: Student must determine the hardware implementation.
- Actions:
    - It must provide MQTT interaction among the client devices
- Required:
    - Topics tree must be defined.

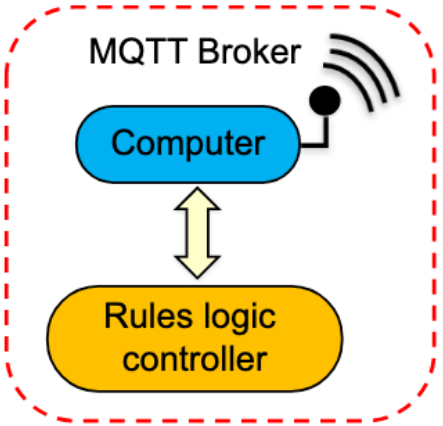
4. e-Game rules monitor

Students must define and implement de logic control to implement the game.




## e-Game rules monitor



## Deliverables:

A final report that includes all the information related to the analysis, development process and final conclusions.

### - Engineering book

The engineering book must contain the project definition:

- *Project Milestones*

  Identify individual goals to be accomplished during the project development. It is necessary to identify their order of execution, priority, dependencies within other milestones, number of working hours, student in charge, etc.

  It can include diagram flows, Gannt chart, etc.

- *For each implemented component:*

  a) A short self-explaining sentence for their goal.
  b) Dedication time (an estimation in hours for the whole team).
  c) A short conclusion on the device development. It could also include any comment related to the team (suggestions, tasks assignments, etc.)

- *e-Game Rules*

  An explanation indicating the e-Game rules, the player and components protocol.

- *Final project conclusions*

  An explanation about the project development, main challenges, obtained results, etc. You can also provide any suggestion related to the project that could improve it as a future work (components, technical specs, user stories, etc.)

### - Technical documentation

Hardware and software development including:

- The wiring schematics for all the components present in the project, so it could be possible to replicate it. The schematic must be understandable and easy to follow.
- MQTT topic tree structure.

* Workgroups can add any other content related to the project that improves the documentation, content, readability, etc.

### - Project Source Code

A **zip file** containing the source code for each component, and if required, any comment that allows to compile and finally to execute the project.

It can be possible to provide a link to the repository you have used in your project (GitHub, GitLab, bitbucket, ….). Remember to give grant access to the repository.

Make sure the code is well structured and commented.

### - Validation Test

At the end of the project, each team member will take a validation test.